Uncertainty Quantification in Medical Image Segmentation: A Comparative Study of Loss Functions Across Datasets

Abstract

Accurate segmentation of medical images is crucial for diagnosis and treatment planning, but quantifying uncertainty in these segmentations is equally vital for reliable clinical use. This study investigates the impact of different loss functions on both segmentation performance and uncertainty estimation, focusing on pancreas and spleen segmentation. Utilizing a U-Net architecture, we trained and evaluated models on two distinct datasets, varying training epochs and learning rates to assess the robustness of our findings. Our results demonstrate that Dice loss consistently outperforms other loss functions in terms of segmentation accuracy and uncertainty calibration across both datasets, even with variations in training regimes. While other loss functions, including log-cosh, focal Tversky, and a hybrid loss, showed potential, their performance and uncertainty estimates were less consistent. This research underscores the importance of careful loss function selection for medical image segmentation, highlighting Dice loss as a robust choice for achieving both high accuracy and reliable uncertainty quantification. Our findings provide valuable insights for developing trustworthy segmentation models for clinical applications.

Keywords: Segmentation, Uncertainty Quantification, Deep Learning, Loss Function, Monte Carlo Dropout

1. Introduction

Accurate medical image segmentation is a cornerstone of modern healthcare, enabling precise diagnosis, treatment planning, and disease monitoring. The pancreas and spleen, in particular, present significant challenges for segmentation due to their complex anatomical shapes, variability among patients, and proximity to other organs. These complexities make reliable segmentation essential yet difficult to achieve, especially using automated methods.

Deep learning models, notably those based on the U-Net architecture[1], have revolutionized medical image segmentation by leveraging convolutional neural networks to capture intricate spatial hierarchies in imaging data. Despite these advancements, a critical aspect that remains under-explored is the quantification of uncertainty in segmentation outputs. In clinical applications, understanding the confidence level of model predictions is vital for ensuring patient safety and building trust in automated systems. Uncertainty quantification[2] allows clinicians to assess the reliability of segmentation results, guiding decision-making processes and highlighting areas that may require further investigation.

The choice of loss function during model training is pivotal not only for achieving high segmentation accuracy but also for influencing the model's ability to estimate uncertainty effectively. Different loss functions[16] can affect the convergence behavior of the training process, the sharpness of segmentation boundaries, and the calibration of confidence scores. While numerous loss functions—such as Dice loss, log-cosh loss, focal Tversky loss, and hybrid losses—have been proposed to enhance segmentation performance, their impact on uncertainty quantification remains less understood.

This study investigates the effect of various loss functions on both the segmentation performance and uncertainty estimation in pancreas and spleen segmentation tasks using a U-Net architecture. By systematically varying training epochs and learning rates, we aim to assess the robustness of each loss function across different training regimes. Our experiments demonstrate that the Dice loss consistently outperforms other loss functions, delivering superior segmentation accuracy and more reliable uncertainty estimates, regardless of the training conditions.

These findings underscore the importance of selecting appropriate loss functions when developing deep learning models for medical image segmentation. By highlighting the Dice loss as a robust choice for achieving both high accuracy and trustworthy uncertainty quantification, our research provides valuable insights that can inform the development of more dependable and clinically applicable segmentation models. This work contributes to bridging the gap between advanced computational methods and their practical deployment in healthcare settings, ultimately aiming to enhance patient outcomes through improved diagnostic tools.

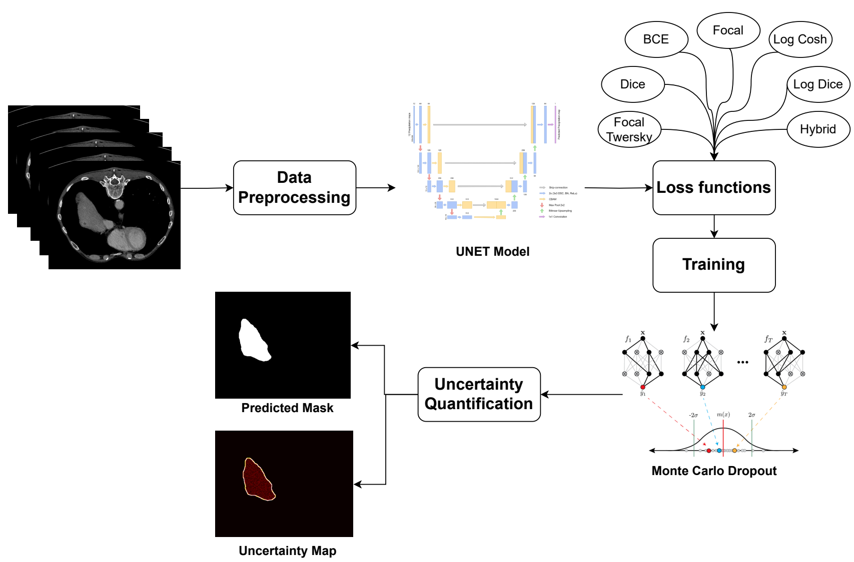


Figure 1: Workflow for Uncertainty Quantification in Medical Image Segmentation. The workflow starts with data collection, followed by preprocessing and training the UNET model with various loss functions (e.g., Dice, BCE, Log Cosh, Log Dice, Tversky, Focal Tversky, and Hybrid). After training, Monte Carlo Dropout is applied for uncertainty estimation, and the results are visualized.

1.1. Datasets

The datasets used in this study were obtained from the Medical Segmentation Decathlon (MSD)[18], a collection curated to promote the development of generalizable algorithms across diverse medical imaging tasks. All data were provided in Neuroimaging Informatics Technology Initiative (NIfTI) format, which is widely adopted for storing medical imaging data due to its ability to handle multidimensional datasets and associated metadata. Access was limited to the training volumes and their corresponding segmentation labels.

1.1.1. Pancreas Dataset

The pancreas dataset focuses on the segmentation of the pancreas and pancreatic tumors from abdominal computed tomography (CT) scans. It comprises 420 three-dimensional (3D) volumes, with 282 volumes designated for training and 138 for testing. The scans were acquired in the portal venous phase and sourced from the Memorial Sloan Kettering Cancer Center. A significant challenge presented by this dataset is the pronounced label imbalance, characterized by large background regions, medium-sized pancreas structures, and small tumor areas. This imbalance poses difficulties for segmentation algorithms, particularly in accurately identifying and delineating the small tumor regions amidst the predominant background.

1.1.2. Spleen Dataset

The spleen dataset consists of CT scans aimed at spleen segmentation, comprising 61 3D volumes with 41 allocated for training and 20 for testing. Also sourced from the Memorial Sloan Kettering Cancer Center, this dataset presents the challenge of a wide range of foreground sizes due to significant variability in spleen size among patients. Such variability requires the segmentation model to be robust to anatomical differences and scale variations, ensuring accurate segmentation across diverse patient anatomies.

1.2. Data Preprocessing

Preprocessing[15] is a critical step in medical image segmentation, aiming to enhance image quality, reduce noise, and standardize data for efficient model training. Proper preprocessing improves model convergence and contributes to more accurate and reliable segmentation outcomes, which is essential for clinical applicability.

1.2.1. Conversion from 3D Volumes to 2D Slices

To reduce computational complexity and memory requirements, the 3D CT volumes were decomposed into two-dimensional (2D) axial slices. This approach simplifies the segmentation task by enabling the use of 2D convolutional neural networks (CNNs), which are less resource-intensive compared to their 3D counterparts.

Each 2D slice was paired with its corresponding segmentation mask, ensuring spatial alignment and consistency. This slicing strategy facilitates more manageable data handling and accelerates the training process without significantly compromising the spatial context needed for accurate segmentation.

1.2.2. Intensity Windowing

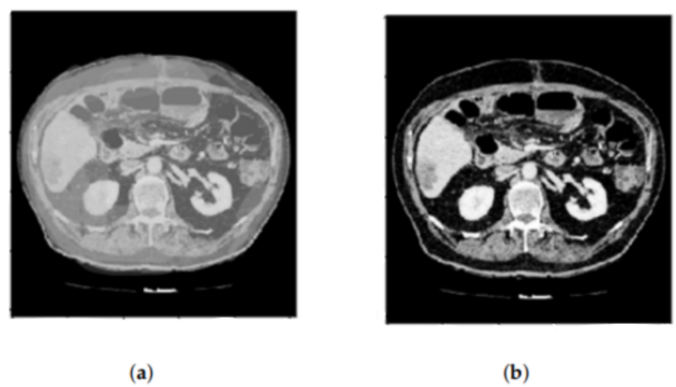


Figure 2: HU windowing of a CT scan slice. The left image shows the original CT slice, while the right image demonstrates the HU (Hounsfield Unit) windowing applied to highlight different tissue densities.

Intensity windowing[15], specifically Hounsfield Unit (HU) windowing, was applied to focus on the relevant intensity ranges that best represent the target organs. CT images encompass a broad spectrum of HU values, and restricting this range enhances the contrast between the organ of interest and surrounding tissues. While the HU value ranges for specific organs can vary depending on the CT scanner and patient-specific factors, windowing was utilized without specifying exact ranges to generalize the method and prevent dependence on scanner-specific parameters. This step is crucial for differentiating the pancreas and spleen from adjacent anatomical structures and mitigating scanner-dependent intensity variations, thereby improving the model's ability to learn meaningful features.

1.2.3. Median Filtering

To further improve image quality, a median filter[15] was applied to each 2D slice. Median filtering is effective for reducing noise while preserving important edge information, which is vital for accurate segmentation. By replacing each pixel value with the median of its neighboring pixels within a specified kernel size (typically 3×3), this non-linear technique diminishes the impact of outliers and random noise, resulting in smoother images with well-preserved anatomical boundaries. The smoothing effect aids the model in focusing on significant structural features rather than being distracted by high-frequency noise.

1.2.4. Data Formatting and Storage

The preprocessed 2D images and their corresponding segmentation masks were saved in Tagged Image File Format (TIFF). TIFF is a versatile format that supports high-resolution images and lossless compression, making it suitable for medical imaging applications where image quality must be preserved. The use of TIFF files facilitated efficient data handling during training and ensured that no degradation of image quality occurred due to compression artifacts. This choice of file format maintained the integrity of the imaging data, which is critical for the performance of deep learning models.

1.3. U-Net Architecture

The U-Net architecture[1] was employed for the segmentation tasks due to its proven effectiveness in biomedical image segmentation. U-Net consists of an encoder-decoder structure with skip connections that facilitate the fusion of low-level and high-level feature maps, enabling precise localization and context understanding. The encoder, or contracting path, captures context through successive convolutional layers with increasing numbers of filters, each followed by a rectified linear unit (ReLU) activation function and a max-pooling operation for downsampling. The decoder, or expanding path, reconstructs the segmentation map using upsampling operations and convolutional layers, with skip connections from the encoder to recover spatial information lost during downsampling.

To optimize the U-Net model for both datasets, adjustments were made to the number of filters and network depth. By varying the number of filters in each convolutional layer, the model's complexity and computational requirements were reduced without compromising performance. This optimization aimed to decrease the carbon footprint associated with model training and inference, aligning with sustainable computing practices. The model maintained sufficient capacity to learn complex features necessary for accurate segmentation while operating more efficiently, demonstrating that environmentally conscious design can coexist with high-performance requirements.

1.4. Loss Functions

Seven different loss functions[16] were explored to evaluate their impact on segmentation performance and uncertainty estimation. Each loss function addresses specific challenges in medical image segmentation, such as class imbalance, boundary precision, and optimization stability. The loss functions employed include Dice Loss, Log-Cosh Loss, Binary Cross-Entropy Loss, Tversky Loss, Focal Tversky Loss, Log Dice Loss, and a Hybrid Loss combining Jaccard Loss, Binary Cross-Entropy Loss, and Structural Similarity Index Measure (SSIM) Loss.

1.4.1. Dice Loss

Dice Loss[16] is derived from the Dice Similarity Coefficient (DSC) and is particularly effective for handling class imbalance by focusing on the overlap between the predicted segmentation P and the ground truth G:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}$$

where p_i is the predicted probability for pixel i, g_i is the ground truth label, ϵ is a small constant to prevent division by zero, and the summations run over all pixels. This loss function emphasizes the correct prediction of the minority class, which is critical in medical segmentation tasks with imbalanced data.

1.4.2. Log-Cosh Loss

Log-Cosh Loss[16] combines the advantages of Mean Squared Error (MSE) and Mean Absolute Error (MAE), providing a smooth and robust alternative:

$$\mathcal{L}_{\text{Log-Cosh}} = \sum_i \log\left(\cosh(p_i - g_i)\right)$$

This loss function is less sensitive to outliers and can improve convergence during training. By smoothly penalizing larger errors, it helps the model to focus on reducing overall prediction discrepancies without being disproportionately influenced by extreme values.

1.4.3. Binary Cross-Entropy Loss

Binary Cross-Entropy (BCE)[16] Loss measures the pixel-wise discrepancy between the predicted probabilities and the ground truth labels:

$$\mathcal{L}_{\text{BCE}} = -\sum_i \left[g_i \log(p_i) + (1 - g_i)\log(1 - p_i)\right]$$

BCE is commonly used in binary classification tasks and is effective for segmentation when combined with other loss functions. It provides a straightforward measure of the difference between predicted probabilities and actual labels, encouraging the model to produce confident and accurate predictions.

1.4.4. Tversky Loss

Tversky Loss[16] generalizes Dice Loss by introducing adjustable parameters to balance false positives and false negatives:

$$\mathcal{L}_{\text{Tversky}} = 1 - \frac{\sum_i p_i g_i + \epsilon}{\sum_i p_i g_i + \alpha \sum_i p_i(1 - g_i) + \beta \sum_i (1 - p_i) g_i + \epsilon}$$

where α and β are weighting factors controlling the penalties for false positives and false negatives, respectively. By adjusting these parameters, the loss function can be tailored to prioritize the reduction of specific types of errors, which is beneficial in medical contexts where the cost of false negatives and false positives may differ significantly.

1.4.5. Focal Tversky Loss

Focal Tversky Loss[[16]] extends Tversky Loss by incorporating a focusing parameter γ, enhancing model training on hard examples:

$$\mathcal{L}_{\text{Focal Tversky}} = \left(1 - \frac{\sum_i p_i g_i + \epsilon}{\sum_i p_i g_i + \alpha \sum_i p_i (1 - g_i) + \beta \sum_i (1 - p_i) g_i + \epsilon}\right)^\gamma$$

This loss function is particularly useful for highly imbalanced datasets, as it focuses the learning process on difficult-to-classify pixels, improving the model's ability to correctly segment small or complex structures.

1.4.6. Log Dice Loss

Log Dice Loss[16] applies a logarithmic transformation to the Dice Coefficient, emphasizing smaller overlap regions and providing stronger gradients when the overlap is minimal:

$$\mathcal{L}_{\text{Log Dice}} = -\log\left(\frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}\right)$$

This transformation makes the loss function more sensitive when the Dice Coefficient is low, encouraging the model to improve in areas where it performs poorly.

1.4.7. Hybrid Loss

The Hybrid Loss[16] combines multiple loss functions to leverage their strengths. In this study, it integrates the Jaccard Loss, Binary Cross-Entropy Loss, and Structural Similarity Index Measure (SSIM) Loss:

$$\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{\text{Jaccard}} + \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{SSIM}}$$

Jaccard Loss

$$\mathcal{L}_{\text{Jaccard}} = 1 - \frac{\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i - \sum_i p_i g_i}$$

SSIM Loss

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(P, G)$$

where $\text{SSIM}(P, G)$ is the structural similarity index computed over the predicted and ground truth images. The inclusion of SSIM Loss encourages the model to consider the structural information within the images, promoting more perceptually accurate segmentations.

1.5. Uncertainty Calculation Using Monte Carlo Dropout Method

Uncertainty quantification[2] is pivotal in medical imaging to evaluate the reliability of model predictions, thereby enhancing clinical decision-making. It provides insights into areas where the model is less confident, indicating the need for cautious interpretation or further analysis.

In this study, uncertainty was estimated using the Monte Carlo Dropout method[4]. During inference, dropout layers with a specified rate (e.g., 0.5) were kept active, introducing stochasticity in the model's predictions. Each input image underwent multiple stochastic forward passes (e.g., $T = 10$), generating a set of predictions. The predictive mean and entropy-based uncertainty were then computed for each pixel.

The entropy-based uncertainty quantifies the randomness in the predicted probability distribution for each pixel:

$$U_i = -\sum_c p_{i,c} \log p_{i,c}$$

where U_i is the uncertainty for pixel i, $p_{i,c}$ is the predicted probability of pixel i belonging to class c, and the summation runs over all classes. Higher entropy values indicate greater uncertainty, signaling that the model's predictions are less reliable in those regions.

By visualizing the uncertainty maps alongside the segmentation results, clinicians can better interpret the model's performance and identify areas that require further investigation or expert review.

1.6. Evaluation Metrics

To comprehensively assess the segmentation models and the reliability of uncertainty estimates, several metrics were employed. Each metric evaluates different aspects of the model's performance and predictive confidence.

1.6.1. Test Dice Coefficient

The Test Dice Coefficient measures the overlap between the predicted segmentation and the ground truth on the test set. It is defined as:

$$\text{Dice} = \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}$$

where:

- p_i is the predicted probability for pixel i.
- g_i is the ground truth label for pixel i.
- ϵ is a small constant to prevent division by zero.

The Dice Coefficient ranges from 0 (no overlap) to 1 (perfect overlap), providing a quantitative measure of segmentation accuracy. A higher Dice score indicates better performance, reflecting the model's ability to accurately segment the target organ.

1.6.2. Uncertainty Mean and Standard Deviation

Uncertainty Mean and Standard Deviation quantify the average predictive uncertainty across all pixels and its variability. These metrics are derived from entropy-based uncertainty calculations using the Monte Carlo Dropout method.

Uncertainty for Each Pixel The entropy-based uncertainty for pixel i is calculated as:

$$U_i = -\sum_c p_{i,c} \log p_{i,c}$$

where $p_{i,c}$ is the predicted probability of pixel i belonging to class c, and the summation runs over all classes.

Uncertainty Mean

$$\text{Uncertainty Mean} = \frac{1}{N}\sum_{i=1}^{N} U_i$$

where N is the total number of pixels. This metric provides an overall measure of the model's predictive uncertainty.

Uncertainty Standard Deviation

$$\text{Uncertainty Std} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(U_i - \text{Uncertainty Mean})^2}$$

This metric assesses the variability of uncertainty across the image, indicating how consistently the model predicts uncertainty.

1.7. Mean Prediction Mean and Standard Deviation

Mean Prediction Mean and Standard Deviation assess the average predicted probabilities and their variability, reflecting the model's confidence and consistency in its predictions.

1.7.1. Mean Prediction for Each Pixel

The mean predicted probability for pixel i across multiple stochastic forward passes is defined as:

$$\bar{p}_i = \frac{1}{T}\sum_{t=1}^{T} p_i^{(t)}$$

where $p_i^{(t)}$ is the predicted probability at the t-th forward pass, and T is the total number of forward passes.

1.7.2. Mean Prediction Mean

The Mean Prediction for all pixels is calculated as:

$$\text{Mean Prediction} = \frac{1}{N}\sum_{i=1}^{N} \bar{p}_i$$

where N is the total number of pixels. This metric indicates the average confidence level of the model across all pixels.

1.7.3. Mean Prediction Standard Deviation

The Mean Prediction Standard Deviation is calculated as:

$$\text{Mean Prediction Std} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (\bar{p}_i - \text{Mean Prediction})^2}$$

This metric measures the variability of the model's predicted probabilities, highlighting regions where the model's confidence fluctuates.

1.7.4. Visualization of Results

To facilitate qualitative assessment, violin plots were generated for both uncertainty and mean predictions across different loss functions. These plots visualize the distribution of uncertainty and predicted probabilities, providing insights into the models' performance and reliability.

1.8. Training Setup

The training configurations were meticulously designed to ensure robust model performance and facilitate a thorough analysis of the loss functions. All experiments were conducted using TensorFlow as the deep learning framework, leveraging its advanced features for building and training neural networks. The models were trained on a workstation equipped with an NVIDIA RTX A5000 graphics card, providing substantial computational power for deep learning tasks.

The datasets were split into training and testing sets with an 80:20 ratio. For the pancreas dataset, 225 volumes were used for training and 57 for testing. For the spleen dataset, 33 volumes were allocated for training and 8 for testing. The Adam optimizer was employed due to its adaptive learning rate capabilities and efficient handling of sparse gradients. An initial learning rate of 0.01 was set for both datasets.

A learning rate scheduler callback[19] was implemented, reducing the learning rate by 90% every four epochs. This strategy prevents vanishing gradients and promotes smooth convergence by allowing the model to make larger updates initially and fine-tune parameters as training progresses. Models were trained for 20 epochs, and an early stopping mechanism monitored the validation Dice Coefficient to prevent overfitting. The best-performing model weights, corresponding to the highest validation Dice Coefficient, were saved for evaluation and deployment.

An ablation study was conducted on the spleen dataset to examine the influence of learning rate variations and extended training on the performance and consistency of the loss functions. Models were trained for 40 epochs with learning rates of 0.1 and 0.001. This exploration aimed to study the nature and consistency of the loss functions under different training regimes, providing deeper insights into how hyperparameter settings affect model performance.

2. Results

2.1. Tabular coloumns

Table 1: Segmentation performance on the Pancreas dataset using different loss functions. The table reports the Dice Coefficient (DCE), Uncertainty Mean, and Uncertainty Standard Deviation across models trained with U-Net architecture.

Loss Function	Dice Coefficient (DCE)	Uncertainty Mean	Uncertainty Standard Deviation
Log Cosh Loss	0.4747	0.008646	0.002048
Dice Loss	**0.6838**	**0.000446**	**0.000732**
Focal Tversky Loss	0.6699	0.000542	0.000867
Tversky Loss	0.6784	0.000555	0.000878
BCE Loss	0.6270	0.001356	0.001828
Log Dice Loss	0.6681	0.000424	0.000690
Hybrid Loss	0.5680	0.000704	0.000969

Table 2: Segmentation performance on the Spleen dataset using different loss functions. The table reports the Dice Coefficient (DCE), Uncertainty Mean, and Uncertainty Standard Deviation across models trained with U-Net architecture.

Loss Function	Dice Coefficient (DCE)	Uncertainty Mean	Uncertainty Standard Deviation
Log Cosh Loss	0.4641	0.036197	0.001626
Dice Loss	**0.8474**	**0.000556**	**0.000798**
Focal Tversky Loss	**0.8954**	**0.000350**	**0.000578**
Tversky Loss	0.8378	0.000598	0.000861
BCE Loss	0.7928	0.005005	0.001539
Log Dice Loss	0.8334	0.000633	0.000791
Hybrid Loss	0.8453	0.000501	0.000703

Table 3: Ablation study on the Spleen dataset with a learning rate of 1×10^{-1} for 40 epochs. Dice Coefficient (DCE), Uncertainty Mean, Uncertainty Standard Deviation, Mean Prediction, and Mean Prediction Standard Deviation are reported. We include the confidence metrics (Mean Prediction and its Standard Deviation) to show that the Dice Loss consistently performs better, indicating higher accuracy and confidence compared to other loss functions.

Loss Function	DCE	Uncertainty Mean	Uncertainty SD	Mean Prediction	Mean Prediction SD
Log Cosh Loss	0.00002	0.000033	0.000015	0.000003	0.000001
Dice Loss	**0.8698**	**0.000327**	**0.000553**	**0.0040**	**0.0086**
Focal Tversky Loss	0.8185	0.000726	0.001031	0.0055	0.0095
Tversky Loss	0.1704	0.004741	0.005464	0.0050	0.0344
BCE Loss	0.0654	0.013086	0.009289	0.0041	0.0036
Log Dice Loss	0.7732	0.000636	0.000817	0.0045	0.0080
Hybrid Loss	0.8181	0.000908	0.001211	0.0049	0.0088

Table 4: Ablation study on the Spleen dataset with a learning rate of 1×10^{-3} for 40 epochs. Dice Coefficient (DCE), Uncertainty Mean, Uncertainty Standard Deviation, Mean Prediction, and Mean Prediction Standard Deviation are reported. The inclusion of the confidence metrics (Mean Prediction and its Standard Deviation) further supports that Dice Loss offers superior performance, achieving higher accuracy and confidence compared to other loss functions.

Loss Function	DCE	Uncertainty Mean	Uncertainty SD	Mean Prediction	Mean Prediction SD
Log Cosh Loss	0.1852	0.111076	0.000765	0.022004	0.008392
Dice Loss	**0.8726**	**0.001892**	**0.000432**	**0.004636**	**0.008829**
Focal Tversky Loss	0.8432	0.002733	0.000663	0.005209	0.009439
Tversky Loss	0.7867	0.006703	0.000724	0.005831	0.009353
BCE Loss	0.2822	0.071881	0.001007	0.022004	0.008392
Log Dice Loss	0.8307	0.005613	0.000521	0.004984	0.008756
Hybrid Loss	0.7946	0.006085	0.000662	0.005754	0.009247

2.2. Training Plots for Different Datasets Under Various Training Regimes

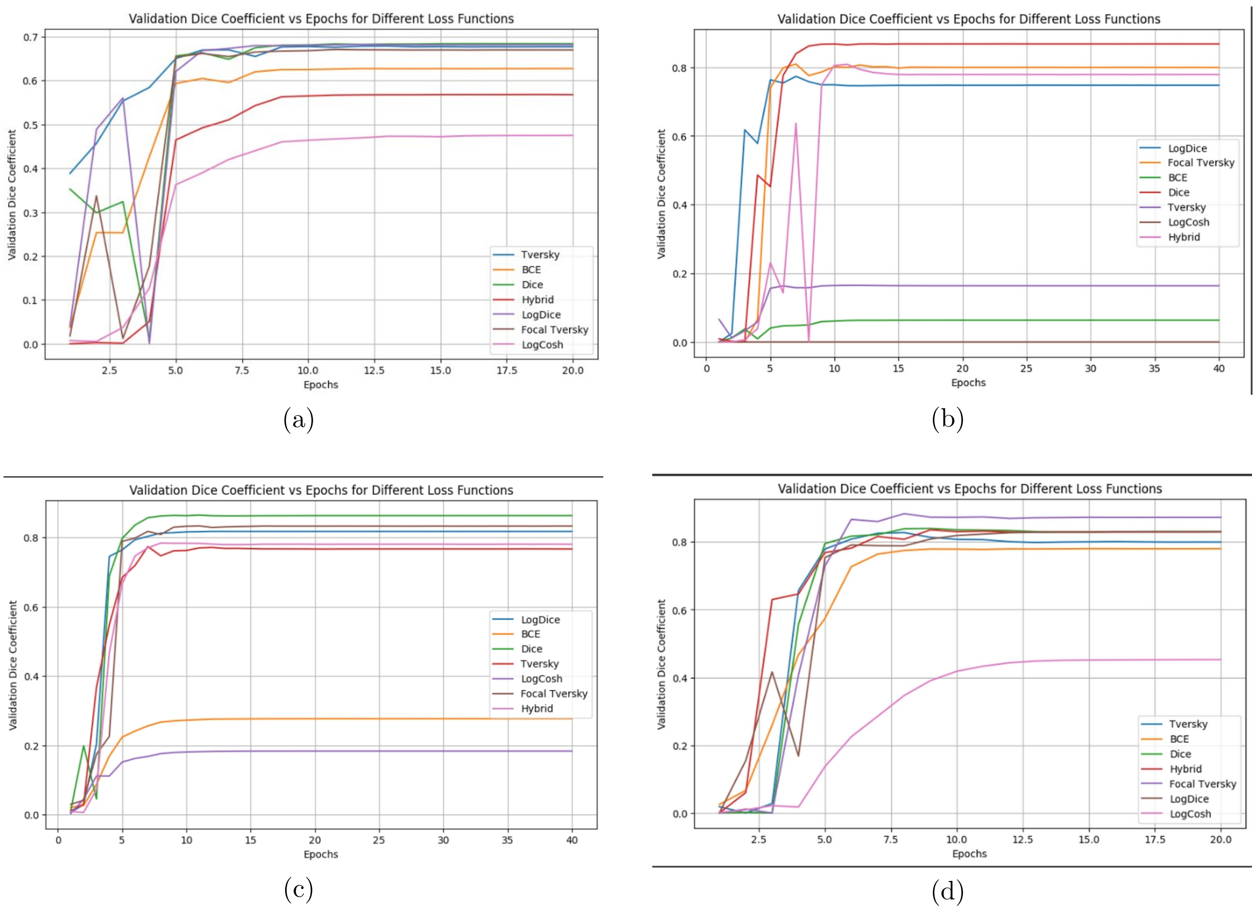
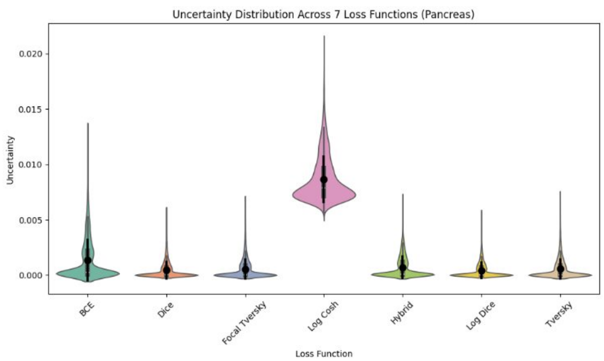
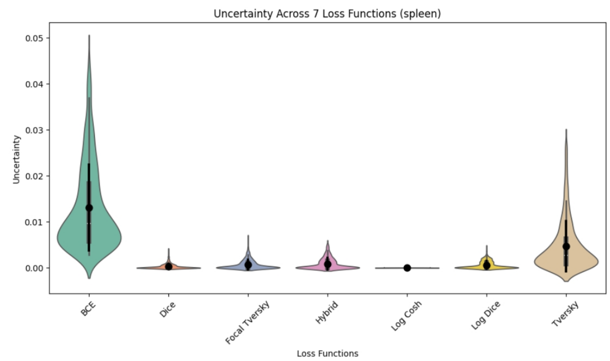


Figure 3: Validation Dice Coefficient plots for different datasets under varying training regimes. (a) shows the training plot for the Pancreas dataset with different loss functions. (b) shows the training plot for the Spleen dataset with a learning rate of 1×10^{-1}. (c) shows the training plot for the Spleen dataset with a learning rate of 1×10^{-3}. (d) shows the training plot for the Spleen dataset over 20 epochs with different loss functions.

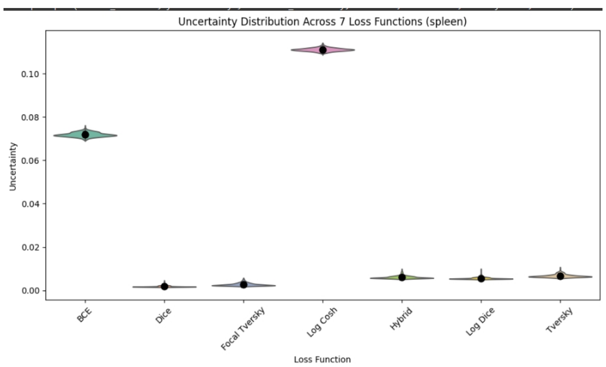
2.3. Uncertainty Violin Plots for Different Datasets Under Various Training Regimes



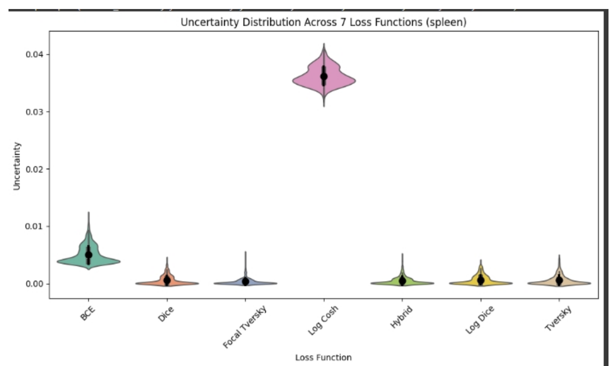
(a)



(b)



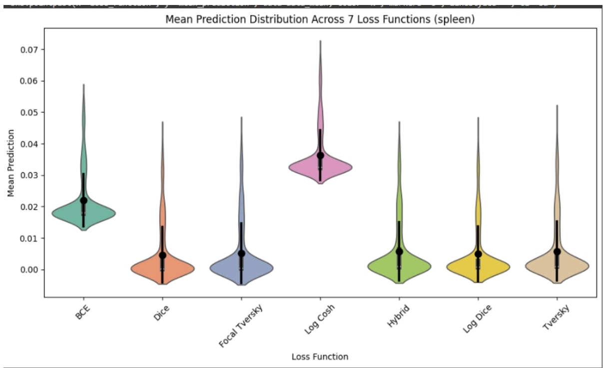
(c)



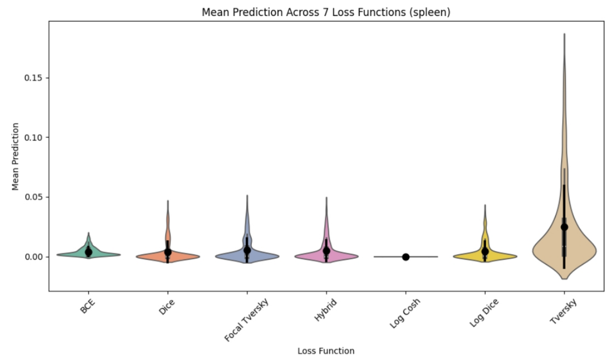
(d)

Figure 4: Uncertainty violin plots for different datasets under varying training regimes. (a) shows the uncertainty violin plot for the Pancreas dataset. (b) shows the uncertainty violin plot for the Spleen dataset with a learning rate of 1×10^{-1}. (c) shows the uncertainty violin plot for the Spleen dataset with a learning rate of 1×10^{-3}. (d) shows the uncertainty violin plot for the Spleen dataset over 20 epochs.

2.4. Confidence Plots for Spleen Dataset with Different Learning Rates



*(a)



*(b)

Figure 5: Confidence plots for the Spleen dataset with different learning rates. (a) shows the confidence plot for the Spleen dataset with a learning rate of 1×10^{-3}. (b) shows the confidence plot for the Spleen dataset with a learning rate of 1×10^{-1}. These plots highlight the model's confidence in its predictions (measured by the Mean Prediction), illustrating that the Dice loss function consistently performs well. The high confidence levels observed for the Dice loss suggest that it not only achieves better segmentation accuracy but also provides more reliable and consistent predictions compared to other loss functions. This makes Dice a strong candidate for tasks requiring both accuracy and prediction certainty.

2.5. loss function predictions

2.5.1. pancreas dataset random sample slice

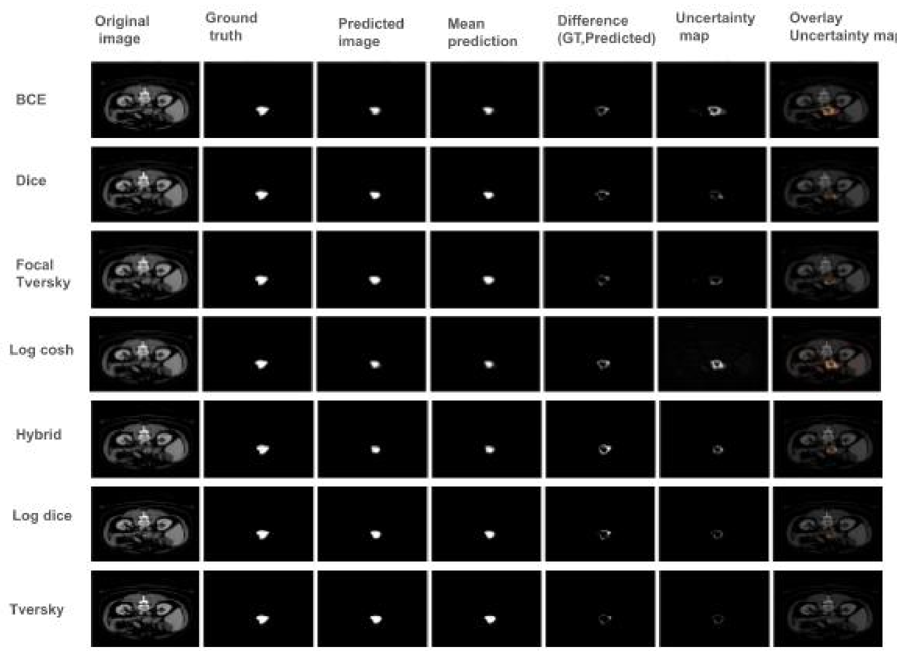


Figure 6: Final visualization results for segmentation using different loss functions on a random slice in pancreas dataset. The rows represent the performance of different loss functions (BCE, Tversky, Dice, Focal Tversky, Hybrid, Log Dice, Log Cosh) on a medical image segmentation task. Columns display the original image, ground truth, predicted segmentation, mean prediction, the difference between ground truth and predicted segmentation, uncertainty map, and an overlay of the uncertainty map on the original image.

2.5.2. spleen dataset random sample slice

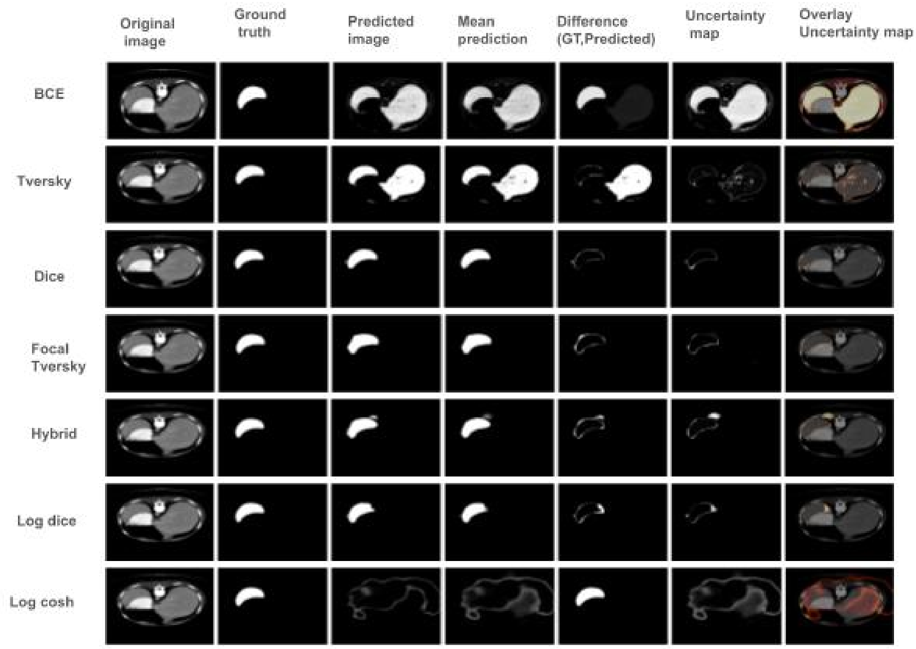


Figure 7: Final visualization results for segmentation using different loss functions on a random slice in spleen dataset. The rows represent the performance of different loss functions (BCE, Tversky, Dice, Focal Tversky, Hybrid, Log Dice, Log Cosh) on a medical image segmentation task. Columns display the original image, ground truth, predicted segmentation, mean prediction, the difference between ground truth and predicted segmentation, uncertainty map, and an overlay of the uncertainty map on the original image.

3. Discussion

The results from our experiments consistently demonstrate the superior performance of the Dice loss function across different datasets and training regimes. This finding is corroborated by the segmentation accuracy, as well as the confidence levels and uncertainty measures of the model's predictions.

3.1. Performance in Segmentation Accuracy

From the segmentation accuracy metrics provided in Tables 1 and 2, it is evident that Dice loss consistently achieves the highest Dice Coefficient (DCE) across both the pancreas and spleen datasets. In contrast, other loss functions, such as Binary Cross-Entropy (BCE) and Tversky, demonstrate lower DCE values, indicating suboptimal overlap between predicted and ground-truth segmentation masks. The performance gap between Dice and these loss functions becomes particularly noticeable when evaluated under different learning rates and training epochs, as illustrated by the validation Dice Coefficient plots (Figure 3).

3.2. Uncertainty Analysis

The violin plots (Figure 4) further highlight the superiority of Dice loss in terms of uncertainty quantification. Models trained with Dice loss exhibit lower uncertainty means and standard deviations compared to other loss functions. This suggests that the predictions made by Dice-optimized models are not only accurate but also more reliable and consistent. In contrast, loss functions such as BCE and Log Cosh display higher uncertainty values, indicating greater variability in their predictions.

Notably, the Focal Tversky loss also performs relatively well in some instances but tends to produce slightly higher uncertainty and variability compared to Dice, as observed in the spleen dataset experiments with different learning rates. This may be attributed to the nature of Focal Tversky loss, which emphasizes difficult-to-segment regions, potentially increasing prediction variability in less ambiguous regions.

3.3. Confidence and Mean Prediction

The confidence plots (Figure 5) offer a deeper understanding of how well the model, trained with different loss functions, generalizes its predictions. Specifically, the Mean Prediction and its Standard Deviation provide insights into the model's confidence in its output. Dice loss, in both learning rate regimes (1×10^{-1} and 1×10^{-3}), exhibits the highest confidence levels with relatively low prediction variability, indicating that it provides not only accurate but also stable and reliable predictions. This is crucial for medical image segmentation tasks where confidence in the predicted boundaries is as important as the segmentation accuracy itself.

Other loss functions, particularly BCE and Hybrid loss, exhibit higher standard deviations in Mean Prediction, indicating that their predictions fluctuate more across multiple stochastic forward passes. This variability can potentially lead to uncertainty in clinical decisions, making Dice loss a more suitable choice for tasks where both accuracy and confidence are required.

3.4. General Observations

Throughout the experiments, the Dice loss function consistently emerges as the most robust option across different settings, outperforming other loss functions not only in accuracy but also in terms of lower uncertainty and higher confidence. The consistent performance of Dice loss across different datasets, learning rates, and epochs suggests that it can generalize well to various medical segmentation tasks, providing reliable outcomes in both well-defined and ambiguous segmentation cases.

While other loss functions such as Focal Tversky and Log Dice perform relatively well in certain instances, they often fall short in terms of confidence and prediction stability, which are crucial factors in sensitive applications like medical diagnostics. This robustness of Dice loss can be particularly useful in clinical workflows where decision-making relies on both the accuracy and reliability of predictions.

4. Conclusion

This study provides a comprehensive analysis of various loss functions in the context of uncertainty quantification for pancreas and spleen segmentation using the U-Net architecture. Across both datasets and under different training regimes—whether varying the number of epochs or the learning rate—Dice Loss consistently emerged as the most robust performer. Notably, the Dice Loss achieved the highest Dice Coefficient alongside low uncertainty means, demonstrating superior accuracy and reliability in predictive confidence. This consistent performance underscores the versatility and robustness of the Dice Loss, making it a strong candidate for clinical applications where both accuracy and confidence in segmentation are paramount.

In contrast, other loss functions, including Log-Cosh, Focal Tversky, and Hybrid Loss, displayed greater variability in both segmentation performance and uncertainty calibration. While these functions showed potential in specific configurations, they lacked the consistency exhibited by Dice Loss across a range of learning rates and epoch settings.

The findings of this study highlight the critical role of loss function selection in developing high-performing segmentation models for medical imaging. By reliably balancing segmentation accuracy with low predictive uncertainty, Dice Loss presents a compelling choice for applications requiring dependable automated segmentation. Furthermore, the ability of the model to maintain high performance despite adjustments in hyperparameters suggests its robustness and suitability for diverse clinical environments. Future work should explore the integration of uncertainty measures into clinical workflows, enhancing trust in automated decision support systems.

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